WORKSHOP MANAGEMENT SYSTEM (WMS) - POSTGRESQL 18

Entity Relationship Diagram (ERD) - Consolidated 6-Table Architecture

Strict Limit: Exactly 6 Tables | Base: Malaysia (RM / 6% SST) | Security: BCrypt | Date: 13 Nov 2026, 19:17

1. CONSOLIDATION ARCHITECTURE (9 TABLES -> EXACTLY 6 NORMALIZED TABLES)

Consolidated Table	Source Entities	Consolidation Strategy & Relational Role
1. customers	customers	Primary client profiles and contact information for vehicle ownership.
2. vehicles	vehicles (no VIN)	Fleet registry without VIN requirement for rapid counter onboarding.
3. staff	staff	Workshop accounts, role authorization, and BCrypt encrypted passwords.
4. catalog_items	service_types + parts	Unified catalog with item_type ('SERVICE'/'PART'), stock, and maintenance intervals.
5. job_orders	job_orders + invoices	Unified work order lifecycle merged with 1:1 billing, tax (6% SST), and payment.
6. job_items	job_services + job_parts	Unified line items table referencing job_orders & catalog_items by item_type.

2. RELATIONAL CARDINALITY & FOREIGN KEY CONSTRAINTS

Relationship	Cardinality	Foreign Key Definition	Integrity Rule
customers -> vehicles	1 to 0..N	vehicles.customer_id -> customers.customer_id	**ON DELETE CASCADE**
vehicles -> job_orders	1 to 0..N	job_orders.vehicle_id -> vehicles.vehicle_id	**RESTRICT**
staff -> job_orders (advisor)	1 to 0..N	job_orders.advisor_id -> staff.staff_id	**RESTRICT**
staff -> job_orders (mechanic)	1 to 0..N	job_orders.mechanic_id -> staff.staff_id	**SET NULL / RESTRICT**
job_orders -> job_items	1 to 0..N	job_items.job_id -> job_orders.job_id	**ON DELETE CASCADE**
catalog_items -> job_items	1 to 0..N	job_items.item_id -> catalog_items.item_id	**RESTRICT**

3. TABLE SCHEMAS & COLUMN SPECIFICATIONS

1. customers (Client Registry)

customer_id	BIGSERIAL (PK)	Unique customer ID
full_name	VARCHAR(150)	Customer legal name
phone	VARCHAR(30)	Contact phone (Index)
email, address	VARCHAR	Optional contact details
created_at	TIMESTAMP	Registration timestamp

2. vehicles (Fleet Registry - No VIN)

vehicle_id	BIGSERIAL (PK)	Unique vehicle ID
customer_id	BIGINT (FK)	Owner (Cascade)
plate_no	VARCHAR(20) (UK)	Registration number
make, model, year	VARCHAR / INT	Car specifications
current_mileage	INT	Intake odometer (km)

3. staff (Personnel & Roles)

staff_id	BIGSERIAL (PK)	Staff identifier
username	VARCHAR(50) (UK)	Login username
password_hash	VARCHAR(255)	BCrypt salted hash
role	VARCHAR(30)	COUNTER | MECHANIC | MANAGE
active, phone	BOOL / VARCHAR	Status flag & phone

4. catalog_items (Unified Catalog)

item_id	BIGSERIAL (PK)	Catalog identifier
item_type	VARCHAR(20)	'SERVICE' or 'PART'
item_code	VARCHAR(60) (UK)	SKU / Package Code
unit_price, cost_price	DECIMAL(10,2)	Price in RM
stock_qty, interval_km	INT	Inventory / Service km

5. job_orders (Consolidated Work Order Lifecycle & 1:1 Invoicing)

job_id (PK)	BIGSERIAL	Work order & invoice identifier
vehicle_id, advisor_id, mechanic_id (FKs)	BIGINT	Vehicle, intake advisor, and assigned technician bay
date_in, date_out, mileage_in	TIMESTAMP / INT	Vehicle intake, completion timestamps, and odometer
complaint, diagnosis, status	VARCHAR	Intake issue, findings, and ticket status (PENDING..INVOICED)
next_service_date, next_service_km	DATE / INT	Projected maintenance recommendations (+6 mos / +10,000 km)
invoice_date, subtotal, tax, discount, total	DATE / DECIMAL	Consolidated billing fields: 6% SST, discounts, total in RM
payment_status, payment_method, paid_at	VARCHAR / TIMESTAMP	Settlement state (PENDING / PAID), method (CASH/CARD/DUITNOW)

6. job_items (Unified Services Rendered & Parts Consumed)

job_item_id (PK)	BIGSERIAL	Line item identifier
job_id (FK), item_id (FK)	BIGINT	Linked work order (Cascade) and catalog reference
item_type, quantity, unit_price	VARCHAR / INT / DECIMAL	Differentiator ('SERVICE'/'PART'), quantity, and RM rate
notes	VARCHAR(255)	Mechanic remarks or installation observations